From a new patient to a plan the team can review

Five illustrated user stories

ML estimates inform scheduling; clinicians confirm dates; planners review changes. Mesa provides separate what-if scenarios.

01 **Give a new patient two explained dates**

Surgeon

02 **Confirm a date and reserve the stay**

Surgeon / bed manager

03 **Plan a cohort, then publish proposals**

Operating-room planner

04 **Respond to a change in room availability**

Operating-room planner

05 **Test the impact of longer hospital stays**

Planner / research team

DEMO CONTEXT

Two rooms, three conventional beds, eight pending requests and four confirmed bookings after the consultation example. All patient references and clinical cases in the screenshots are synthetic. Trained duration and LOS models were enabled.

A/B dates use the fast individual scheduler; cohort planning calls the selected engine. Mesa uses simplified scenarios. Model estimates still need validation and calibration for the web input fields.

Give a new patient two explained dates

As a surgeon, I want room-time and length-of-stay estimates from the consultation details, so that I can discuss feasible dates with the patient.

Actions: Open Nouvelle demande; enter a pseudonym, birth year, sex, surgeon, diagnosis, CCAM code, procedure type and requested date window; choose Créer et proposer des dates.

Result: DEMO-NOUVEAU receives Monday 19 October and Friday 23 October. Each option explains room load, bed availability throughout the predicted stay, and the 10% vacation margin.



Two possible dates with capacity explanations and the trained-model prediction.

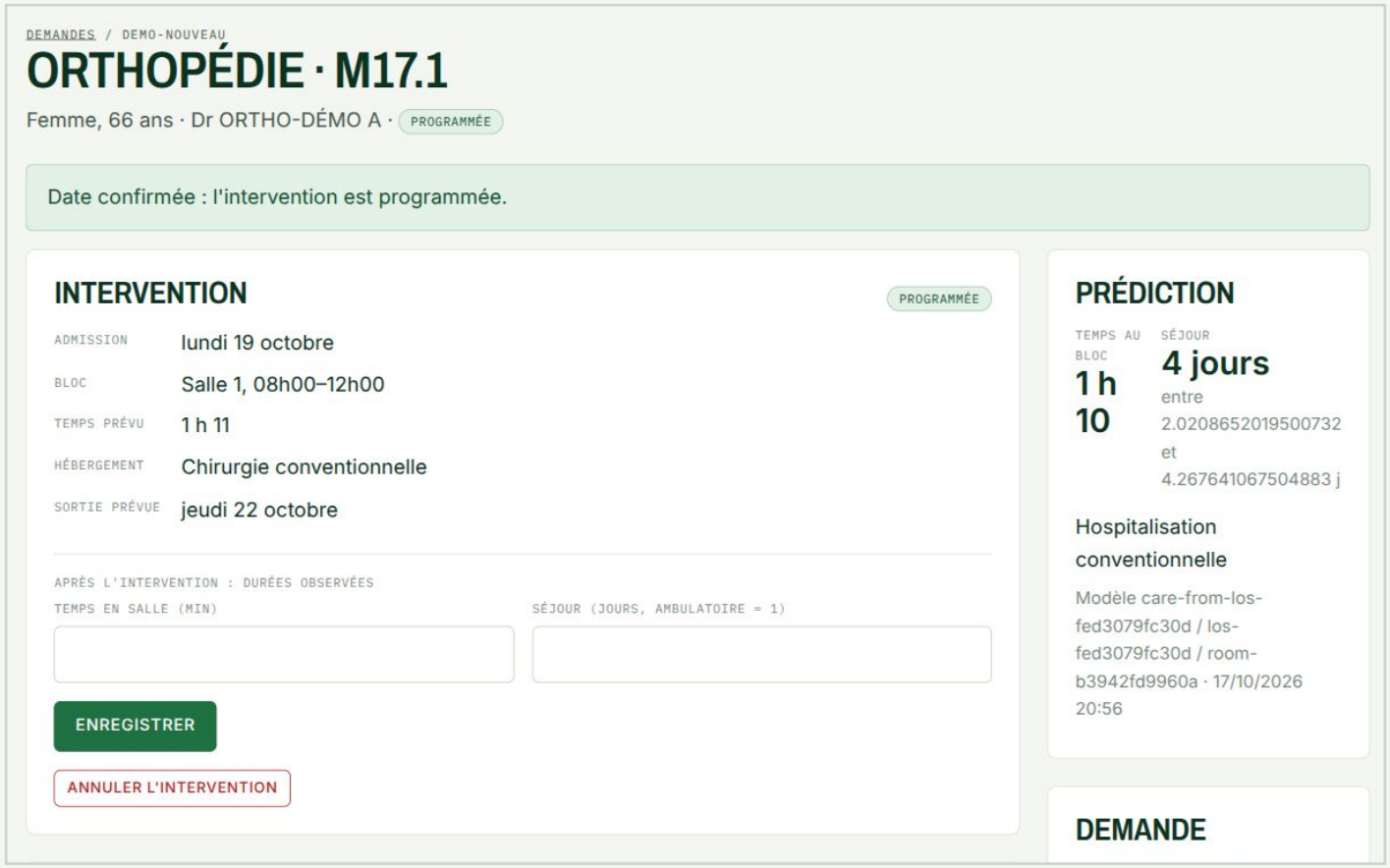
The prediction is about 70 minutes and a four-day stay. A date proposal uses 71 rounded minutes. These are estimates of room occupancy, including more than the surgical act alone.

Confirm a date and reserve the stay

As a surgeon, I want to confirm one proposed date, so that the room capacity and beds for the predicted stay are reserved together.

Actions: Review Date A and select Confirmer cette date. Then open Lits to inspect the resulting occupancy forecast.

Result: The request becomes Programmée for 19 October, with a predicted last occupied day of 22 October. In this demo, conventional occupancy rises from one to two beds on those four days.



The confirmed intervention, its predicted stay and fields for later observed outcomes.

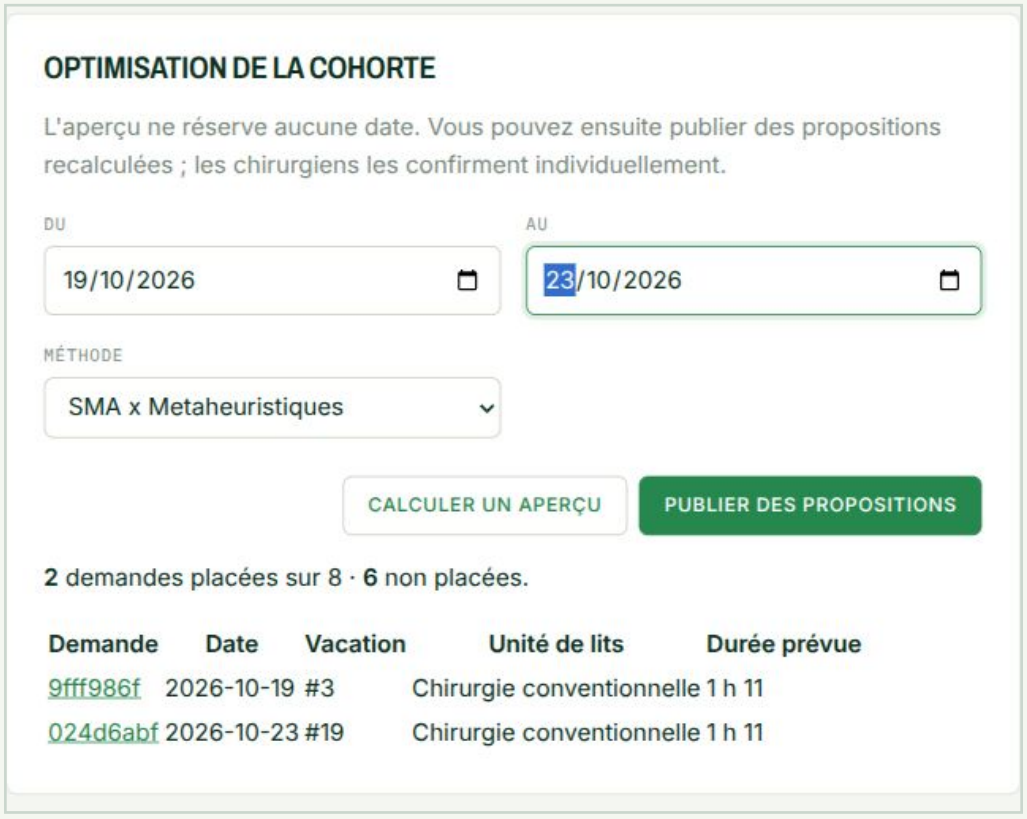
Confirmation rechecks current capacity. A proposal by itself creates no reservation. Observed room time and stay can be recorded after surgery; recording them does not automatically retrain the models.

Plan a cohort, then publish proposals

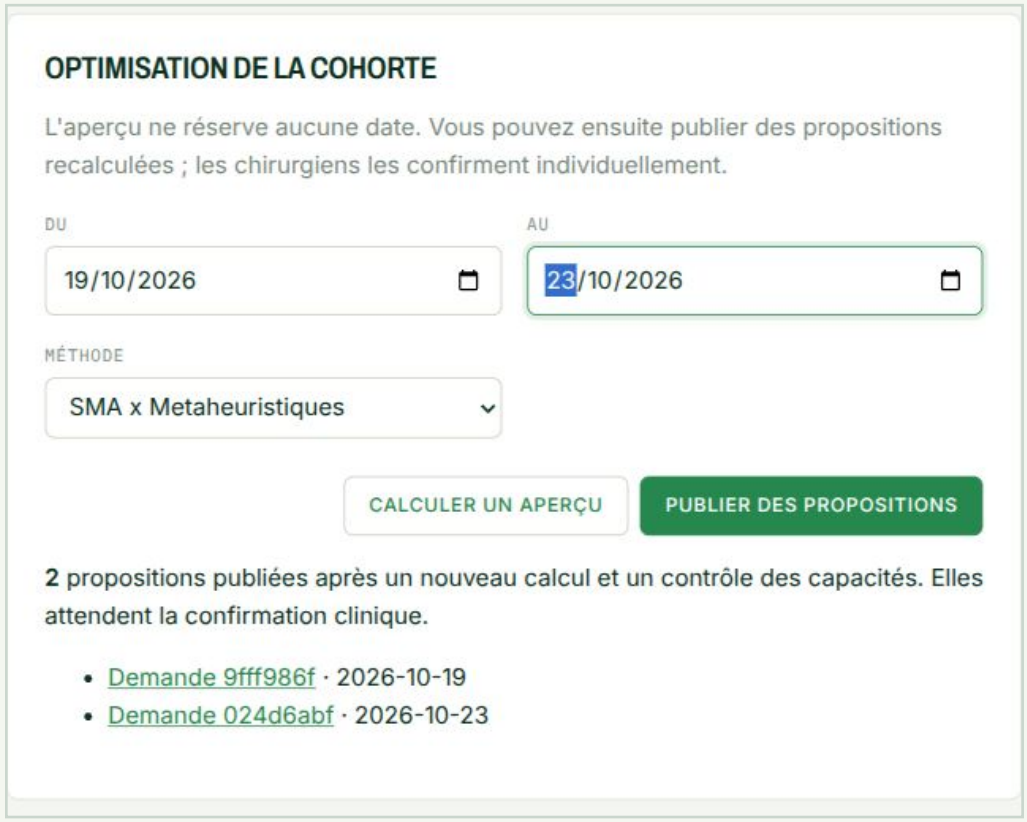
As a planner, I want to optimize pending requests together, so that shared room and bed constraints are considered across the cohort.

Actions: In Bloc, set 19-23 October under Optimisation de la cohorte; choose SMA x Metaheuristiques; calculate an overview, then select Publier des propositions.

Result: The combined engine places two of eight pending requests; six remain unassigned. Publication recalculates and checks capacity, then publishes two proposals awaiting clinical confirmation.



Preview: two assignments and six unassigned requests.



Publication: two proposals, each awaiting clinical confirmation.

Confirmed cases are preserved. Unassigned cases remain visible. Published proposals are still provisional and do not reserve beds or room time.

Respond to a change in room availability

As a planner, I want resource changes to trigger a review of pending proposals and highlight affected confirmed cases, so that the team can respond to a disruption.

Actions: In Ressources, deactivate Salle 1. Return to Bloc, inspect Interventions confirmées à revoir and select Actualiser les propositions.

Result: Four confirmed interventions are flagged because their room is unavailable. The refresh completes and updates two pending proposals; the queue returns to zero pending changes.



The four affected confirmed interventions require review.



The completed refresh reports two updated proposals.

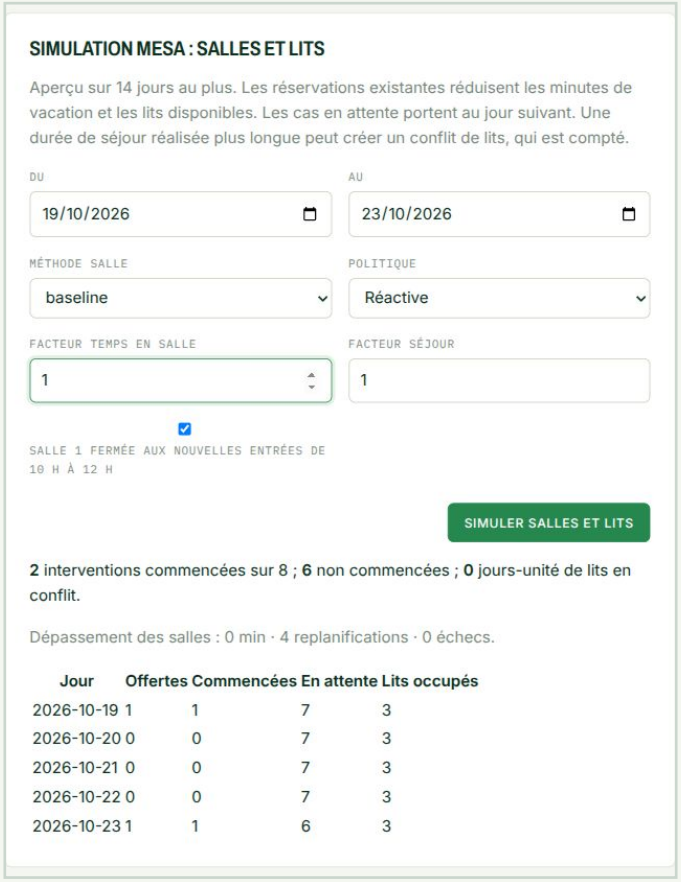
Confirmed reservations remain in place for team review. The refresh does not automatically relocate confirmed operations. This demo uses manual refresh; a background worker is optional.

Test the impact of longer hospital stays

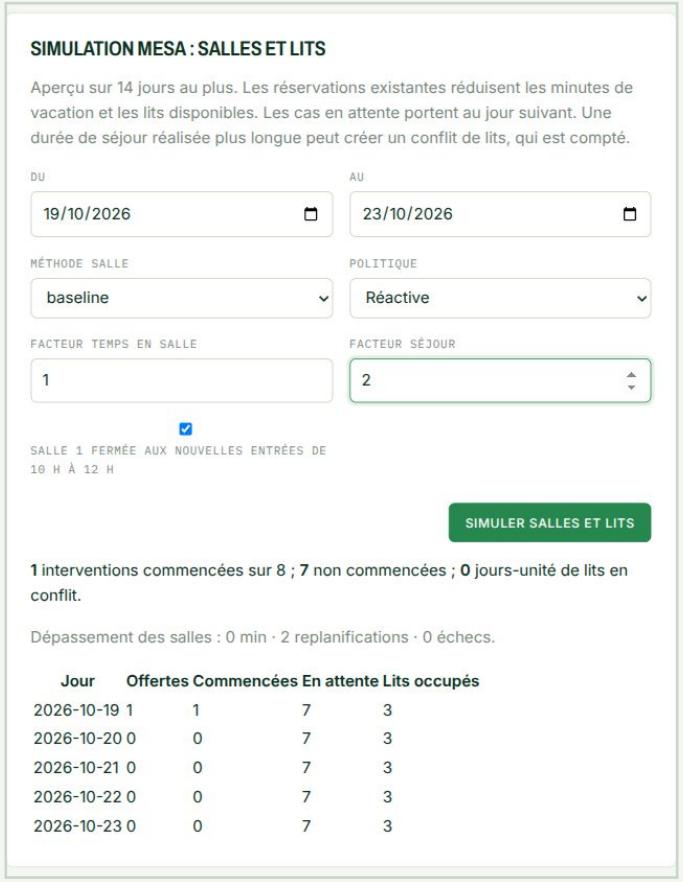
As a planner, I want to test a hypothetical scenario before changing the live plan, so that I can see whether beds or room capacity limit admissions.

Actions: Use Simulation Mesa : salles et lits for 19-23 October. Keep baseline, Réactive, room factor 1 and the 10:00-12:00 closure. Compare stay factor 1 with stay factor 2.

Result: With the same eight pending cases and four confirmed bookings, factor 1 starts two cases and leaves six unstarted. Factor 2 starts one and leaves seven unstarted. Both report zero bed-conflict days and zero solver failures.



Stay factor 1: two started; six unstarted; four replans.



Stay factor 2: one started; seven unstarted; two replans.

These are hypothetical execution durations derived from predictions. They are not observed patient outcomes, oracle forecasts, or a comparison proving that one solver is best. Simulation creates no bookings.